# HVAC system

create data-Flow diagram for fanSpeedLevel, boost, newSpeed

No loops, in Constructor, one branch for validation of `fanSpeedLevel` testing that creation of super component works correctly, affecting file `SmartDevice.java` rest can be checked by statement coverage

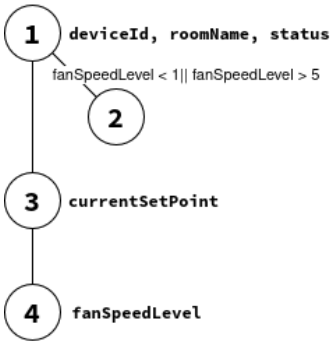


Figure 1: CFG of the Constructor

in adjustFanSpeed, one initialization, three branches, two validating one default outcome as assignment. The Data-Flow for boost is initialization as parameter, usage in statement 1, for newSpeed is initialization in statement one, predicate usage in statement two and 3 and usage in Statement 4, for fanSpeedLevel computational usage in statement 1, assignment in statement 2-4

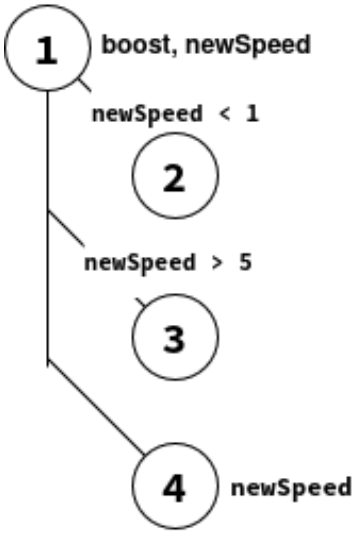


Figure 2: CFG of the adjustFanSpeed method

## White-Box-Tests-Constructor

**Branch-Test 1**
new speed below 1

**Branch-Test 2**
new speed above 5

**Statement-Test 3**
new speed in valid range, get currentSetPoint

**Statement-Test 4**
new speed in valid range, get fanSpeedLevel

## White-box-Tests-adjustFanSpeed

**Branch-Test 1**
boost = −5 Assert getFanSpeedLevel = 1

**Branch-Test 2**
boost = 5 Assert getFanSpeedLevel = 5

**Branch-Test 3**
boost = 0 Assert getFanSpeedLevel = previous getFanSpeedLevel

## Black-Box-Tests

**Black-Box Test 1**
Fan speed increases boost = 1

**Black-Box Test 2**
Fan speed decreases boost = −1

**Black-Box Test 3**
Fan speed stays above 1 boost = −42

**Black-Box Test 4**
Fan speed stays below 5 boost = +42

**Robust-Worst-Case BVA**
only one input variable with no input boundaries, but fanSpeedLevel has boundaries, leading to 7 data-points,

| initial fanSpeedLevel | boost | validity | final fanSpeedLevel |
|---|---|---|---|
| 3 | −3 | `true` | 1 |
| 3 | −2 | `true` | 1 |
| 3 | −1 | `true` | 2 |
| 3 | 0 | `true` | 3 |
| 3 | 1 | `true` | 4 |
| 3 | 2 | `true` | 5 |
| 3 | 3 | `true` | 5 |

**Weak-Robust ECP**
There could be 1 valid class or 5, both versions covered in Robust-Worst-Case BVA